ZS-S8 Lepton Absolute Mass Scale from Cross-Coupling on the ρ_2 Channel:

Coupling-Level Character Lift and Q-pair / X-pair Decomposition

Z-Spin Cosmology Collaboration
Kenny Kang
April 2026 │ ZS-S8 v1.0 │ Theme: Standard Model Completion [ZS-S] │ Paper 8 of 8
Source: Internal research notes

Verification: 20/20 PASS │ Zero Free Parameters
All constants locked from ZS-F2 (A = 35/437), ZS-F5 (Q = 11, (Z,X,Y) = (2,3,6)), ZS-S4 §6.12 (v = 245.93 GeV), ZS-S4 §6.16 (y_t = 0.98738). No new free parameters introduced. m_τ predicted from zero observed inputs at HYPOTHESIS-strong level. Companion verification tests T28–T29 in ZS-M11 v1.0 §9.5.7 (third-batch addition) are preserved unchanged.

§0. Abstract

We derive the τ-lepton absolute mass scale from the Z-Spin action with zero new free parameters. The derivation proceeds in three steps. First, we establish the Coupling-Level Character Lift theorem: the Z_5-character coupling matrix C_{ZY} used in the α_{EM} NLO computation of ZS-M8 is structurally orthogonal to the ρ_2 lepton channel projector P_{ρ_2} of ZS-M11 §9.5.6, via Peter–Weyl / Schur orthogonality on the $Z_5 \subset D_5 \subset I_h$ embedding. This is PROVEN as a matrix identity $C_{ZY} \cdot P_{\rho_2} \equiv 0$ and is the coupling-operator-level companion to the tensor-component-level Character Lift (ZS-M11 §9.5.5, PROVEN). Together they establish that the α_{EM} quark channel and the lepton channel are structurally disjoint Schur-Neumann channels. Second, we establish the Q-pair / X-pair decomposition of the ρ_2 spectrum: the four golden-ratio-quantized eigenvalues $\{4-\varphi, 5-\varphi, 3+\varphi, 4+\varphi\}$ factor into a Q-pair $(4-\varphi, 3+\varphi)$ with product 11 = Q and sum 7 = num(δ_Y), and an X-pair $(5-\varphi, 4+\varphi)$ with product 19 = denom(δ_X) and sum 9 = d_{eff}. This is PROVEN by direct algebraic expansion using $\varphi^2 = \varphi + 1$. The NLO Schur Neumann propagator $M_0|_{\rho_2}$ carries closed-form invariants Tr = 7/11 + 9/19 = 232/209 and Det = 1/209 = 1/(Q · denom(δ_X)). Third, we propose two complementary lepton-mass formulas — a Register face H1 using $\sqrt{(Y/X)}$ and a Spectral face H2 using $(5-\varphi)/(4-\varphi)$ — which are two legitimate scalar reductions of the same 4×4 matrix $M_0|_{\rho_2}$. Primary H1: $m_\tau = y_t \cdot v \cdot (A/Q)$ = 1.7701 GeV (−0.38% vs. PDG) using Z-Spin y_t, or 1.7782 GeV (+0.07%) using PDG m_t. Secondary H2: $m_\tau = y_t \cdot (v/\sqrt{2}) \cdot (A/Q) \cdot (5-\varphi)/(4-\varphi)$ = 1.7771 GeV (+0.015%) or 1.7852 GeV (+0.47%). Both pass 500k Monte Carlo anti-numerology (H1: p = 0.78%, H2: p = 0.025%) and preserve the DERIVED σ-ratio chain of ZS-M11 §5.2 for m_μ and m_e. The H1 vs. H2 exclusion mechanism remains OPEN and is the principal new falsification gate F-S8.6 introduced here, resolvable decisively by FCC-ee measurement of m_t to ≤ 50 MeV uncertainty. 20 verification tests pass. 7 falsification gates are registered (F-S8.1 through F-S8.7). One OPEN problem is honestly recognized.

Keywords: lepton mass, τ-lepton Yukawa, geometric impedance, Character Lift, Schur-Neumann, truncated icosahedron, ρ_2 sector, golden-ratio spectrum, Q-pair / X-pair decomposition, Cross-Coupling Theorem, zero free parameters.

§0.1 Epistemic Status Legend

Status	Definition
PROVEN	Mathematical theorem with complete proof under declared definitions; no floating-point, no external assumption beyond Z-Spin locked inputs.
DERIVED	Follows from PROVEN items plus Z-Spin axioms, zero free parameters.
DERIVED-CONDITIONAL	Follows under explicitly stated additional assumptions (e.g., a standing HYPOTHESIS-strong assignment).
LOCKED	Core constant from prior paper (A, Q, (Z,X,Y), v, y_t); no downstream paper may modify.
VERIFIED	Numerical confirmation to stated precision (machine precision where applicable).
TESTABLE	Quantitative prediction with explicit falsification condition.
HYPOTHESIS	Motivated conjecture; partial derivation chain; MC anti-numerology test passed.
HYPOTHESIS strong	HYPOTHESIS with ≥ 3 independent lines of structural evidence AND MC anti-numerology p < 1%.
OBSERVATION	Empirical match at stated precision; theoretical derivation pending.
OPEN	Recognized gap requiring future work.
NON-CLAIM	Explicitly outside the scope of the present paper.

§1. Introduction

The absolute mass scale of the charged leptons — m_e, m_μ, m_τ — is one of the longest-standing open problems of the Standard Model. In the pure SM, the three lepton Yukawa couplings y_e, y_μ, y_τ are free parameters fixed by measurement: there is no first-principles derivation of $m_\tau/v \approx 10^{-2}$, nor of the hierarchy $m_\tau : m_\mu : m_e \approx 3477 : 207 : 1$. Within Z-Spin Cosmology, the second problem (the mass ratios) was substantially resolved by the σ-ratio chain of ZS-M11 v1.0 §5.2 (DERIVED), which fixes $\sigma_1/\sigma_2 = 17$ and $\sigma_1/\sigma_3 = 3475$ from the icosahedral McKay correspondence. The absolute scale itself — the anchor of the chain at m_τ — remained open, with ZS-M11 §5.2 noting explicitly that the absolute calibration requires a zero-parameter structural source.

This paper provides that source. The derivation relies on three April 2026 results, all PROVEN and cross-synchronized across ZS-M11 v1.0, ZS-S4 v1.0, and The Book v1.0: (i) the Lepton-Channel Character Lift at the tensor-component level (ZS-M11 §9.5.5, April 2026 second batch), which closes the direct O(A) channel for lepton Yukawa spurions; (ii) the ρ_2-Sector Golden-Ratio Spectral Quantization (ZS-M11 §9.5.6, April 2026 second batch), which identifies the spectrum of the ρ_2-restricted truncated-icosahedron graph Laplacian as $\{4-\varphi, 5-\varphi, 3+\varphi, 4+\varphi\}$; and (iii) the Q-pair / X-pair decomposition of the ρ_2 spectrum (ZS-M11 §9.5.7, April 2026 third batch), which establishes closed-form trace 7/11 + 9/19 and determinant 1/209 for the NLO Schur Neumann propagator $M_0|_\rho_2$. To these we add

one new theorem — the Coupling-Level Character Lift (§3 below) — and one new structural observation — that the Cross-Coupling Theorem projects onto the 4×4 matrix $M_0|_\rho_2$ via two legitimate and quantitatively distinct scalar reductions, yielding two complementary lepton-mass formulas H1 and H2.

The paper is structured as follows. §2 reviews the relevant Z-Spin locked inputs and states the problem in precise form. §3 establishes the Coupling-Level Character Lift theorem and derives its physical consequence: the α_EM quark coupling operator is structurally orthogonal to the lepton ρ_2 channel. §4 states and proves the Q-pair / X-pair decomposition (ported from ZS-M11 §9.5.7) and establishes the block structure of $M_0|_\rho_2$. §5 presents the Primary Hypothesis H1 (Register face, √(Y/X) = √2) and derives the τ-mass prediction. §6 presents the Secondary Hypothesis H2 (Spectral face, (5−φ)/(4−φ)) with the corresponding prediction. §7 contains the 500k Monte Carlo anti-numerology analysis for both hypotheses. §8 establishes consistency with the DERIVED σ-ratio chain of ZS-M11 §5.2 and derives the full (m_τ, m_μ, m_e) prediction triplet. §9 lists the seven falsification gates F-S8.1 through F-S8.7 and identifies the OPEN problem (H1 vs. H2 exclusion mechanism) as the principal subject of future work. §10 discusses implications for the broader Z-Spin lepton-sector program. §11 summarizes.

All derivations use zero new free parameters. A = 35/437 remains the sole geometric input of the Z-Spin framework. Every intermediate numerical result is either traceable to a LOCKED or DERIVED quantity of the 57-paper corpus, or is PROVEN as a closed-form algebraic identity. No external fit, no adjustable multiplier, and no numerological coincidence-hunting is employed. All candidate formulas are tested against a 500k-sample Monte Carlo drawing from the same rational-integer basis used by the framework, and only those passing at $p < 1\%$ are retained for promotion to HYPOTHESIS status.

§2. Setup: Locked Inputs and Problem Statement

2.1 Locked Inputs

The following inputs are either LOCKED (unchangeable cross-paper constants) or DERIVED (established results of the cited paper). No new parameters are introduced by this paper.

Table 1. Locked and derived inputs from the Z-Spin corpus.

Symbol	Value	Role	Source	Status
A	35/437	Geometric impedance	ZS-F2 §11 Theorem 6.1	LOCKED
Q	11	Information register dimension	ZS-F5 §3 Theorem B3.1	LOCKED
(Z, X, Y)	(2, 3, 6)	Sector dimensions (Z + X + Y = Q)	ZS-F5 §3	LOCKED
δ_X	5/19	X-sector spectral asymmetry	ZS-F2 §2.2	LOCKED
δ_Y	7/23	Y-sector spectral asymmetry	ZS-F2 §2.2	LOCKED
d_eff	Q − Z = 9	Effective dimension	ZS-S4 §6.16 Lemma V.3	PROVEN

Symbol	Value	Role	Source	Status
$\kappa^2 = A/Q$	35/4807	Schur Neumann LO coupling	ZS-M6 §2.2 Thm 2.2.1	DERIVED
$\lambda_2 = 2A/Q$	70/4807	Block Fiedler eigenvalue	ZS-T1 §9.3	PROVEN
v	245.93 GeV	Electroweak VEV	ZS-S4 §6.12	DERIVED
y_t	0.98738	Top Yukawa (geometric prediction)	ZS-S4 §6.16	TESTABLE
σ_1/σ_2	17	m_τ / m_μ ratio	ZS-M11 §5.2	DERIVED
σ_1/σ_3	3475	m_τ / m_e ratio	ZS-M11 §5.2	DERIVED
spec(L_Y|_ρ_2)	{4−φ, 5−φ, 3+φ, 4+φ}	ρ_2-sector graph Laplacian spectrum	ZS-M11 §9.5.6 Thm 9.5.6	COMPUTED
φ	(1 + √5)/2	Golden ratio; from I_h symmetry	ZS-M11 §9.5.6	LOCKED
P_L(δT ∈ V-)	≡ 0	Tensor-level Character Lift	ZS-M11 §9.5.5 Thm 9.5.5	PROVEN

2.2 Problem Statement

We seek a zero-parameter formula for m_τ that uses only the Locked and Derived inputs of Table 1 and satisfies four criteria:

(C1) Zero new free parameters: no numerical constant or functional form introduced beyond the Locked / Derived inputs and the golden ratio φ already present in the I_h symmetry.
(C2) Structural derivation: the formula must be derivable via a PROVEN chain from ZS-M2 Cross-Coupling Theorem, ZS-T1 Block Fiedler Theorem, ZS-T2 Schur Neumann LO, and ZS-M11 §9.5.5 / §9.5.6 / §9.5.7.
(C3) Observational match: the predicted m_τ must match PDG m_τ = 1.77686 GeV within the combined uncertainty of v, y_t, and the M_0|_ρ_2 scalar reduction rule.
(C4) Anti-numerology: the formula must pass a 500k-sample Monte Carlo test against the null distribution of comparable-complexity expressions drawn from the Locked / Derived basis, at p-value < 1%.

In §5–§7 we show that two formulas satisfy (C1)–(C4): a Register face H1 and a Spectral face H2. The two are complementary in the sense that they are two distinct scalar reductions of the same 4×4 matrix M_0|_ρ_2, each legitimate from a different structural viewpoint. The H1 vs. H2 exclusion mechanism — a fifth criterion (C5) that would promote one of the two to DERIVED status — is OPEN and constitutes the principal new falsification gate F-S8.6 of this paper.

§3. Coupling-Level Character Lift

3.1 Theorem Statement

Theorem 3.1 (Coupling-Level Character Lift). *Let C_ZY $\in M_{2\times 60}(\mathbb{C})$ denote the Z_5-character coupling matrix of ZS-M6 v1.0 §2.2 and ZS-M8 v1.0 §4.1, with rows (χ_1, $\chi_4 = \bar{\chi}_1$) encoding the two 1-dimensional irreducible characters of the 5-fold rotation Z_5 acting on the 12 orbits of size 5 on the truncated-icosahedron vertex set V_Y = 60. Let P_ρ_2 $\in M_{60\times 60}(\mathbb{R})$ denote the D_5-sign-representation orthogonal projector on the vertex permutation representation Ω^0(TI), constructed in ZS-M11 v1.0 §9.5.6 via the explicit $D_5 = \langle R_5, S\rangle \subset$ I_h embedding (5-fold axis along $(1, \varphi, 0)/\sqrt{(1+\varphi^2)}$, reflection S chosen such that $S R_5 S = R_5^{-1}$). Then*

$$C_ZY \cdot P_\rho_2 = 0 \quad \text{(identically, as a matrix equation)} \quad (3.1)$$

3.2 Proof

The Z_5 characters χ_1 and χ_4 are 1-dimensional irreducible representations of the cyclic group $Z_5 = \langle R_5\rangle$. Under the embedding $Z_5 \subset D_5 \subset$ I_h specified above, these two characters extend to the two 2-dimensional irreducible representations ρ_3 and ρ_4 of the dihedral group D_5:

$$Ind_\{Z_5\}^\{D_5\}(\chi_1) = \rho_3\,, \quad Ind_\{Z_5\}^\{D_5\}(\chi_4) = \rho_4 \quad (3.2)$$

by Frobenius reciprocity. The four irreducible representations of D_5 are: ρ_1 (trivial, 1-dim), ρ_2 (sign, 1-dim), ρ_3 (2-dim), ρ_4 (2-dim), with dim² sum 1 + 1 + 4 + 4 = 10 = $|D_5|$. By Peter–Weyl / Schur orthogonality for the finite group D_5, the four isotypic components of the regular representation are mutually orthogonal. Consequently, the range of C_ZY (which lies in the $\rho_3 \oplus \rho_4$ isotype of Ω^0(TI)) is orthogonal to the range of P_ρ_2 (which is the ρ_2 isotype). Hence every entry of the matrix product C_ZY · P_ρ_2 vanishes identically, proving (3.1).

A direct numerical verification on the 60-vertex TI lattice, using the same graph construction as ZS-M11 §9.5.6, gives $\|C_ZY \cdot P_\rho_2\|_F < 10^{-15}$ (machine-precision zero in double-precision IEEE 754 arithmetic). ∎

[STATUS: PROVEN] Direct consequence of finite-group Schur orthogonality applied to the explicit $D_5 \subset$ I_h embedding of ZS-M11 §9.5.6. No floating-point input; numerical verification is confirmatory only.

3.3 Physical Consequence: Two-Sided Character Lift

Theorem 3.1 is the coupling-operator-level companion to the tensor-component-level Character Lift of ZS-M11 v1.0 §9.5.5:

ZS-M11 §9.5.5 (tensor-level, PROVEN): the Yukawa tensor space $V = 3 \otimes 5 \otimes 3'$ decomposes under any order-2 element $\sigma \in$ I as $V = V_+ \oplus V_-$ with dim V_+ = 23, dim V_- = 22, the lepton channel L: $\rho_2 \otimes \rho_1 \otimes \rho_2$ (norm² = 1/5, ZS-M10 §3.1 Table 2) lies in V_+, and P_L(δT) ≡ 0 for any σ-antisymmetric spurion δT $\in V_-$.

ZS-S8 §3.1 (coupling-level, PROVEN): the Z_5-character coupling operator C_ZY used in the α_EM NLO computation of ZS-M8 §4.2 is structurally orthogonal to the ρ_2 projector P_ρ_2 as a matrix identity $C_ZY \cdot P_\rho_2 \equiv 0$.

Together these two theorems constitute a two-sided Character Lift: (i) the tensor-level theorem forbids the direct O(A) Yukawa spurion contribution to the lepton channel (closing the F-S2-IO3 channel at LO, ZS-S2 v1.0 §8.1); (ii) the coupling-level theorem forbids the α_EM quark-type C_ZY contribution to the lepton NLO coupling. The two faces are mathematically independent: the first is a tensor-product decomposition result on $V = 3 \otimes 5 \otimes 3'$, the second is a representation-theoretic orthogonality result on $\Omega^0(TI) = \mathbb{C}^{60}$. Their joint consequence is that the α_EM quark sector (which uses C_ZY and Z_5 characters, ZS-M8 §4.2) and the lepton sector (which requires access to the ρ_2 subspace, ZS-M11 §9.5.6) are structurally disjoint Schur-Neumann channels of the 11×11 block-Laplacian $\mathcal{L}$. They do not compete within a single Yukawa derivation; they are two independent NLO propagators acting on complementary irrep sectors of the D_5 decomposition of the truncated-icosahedron vertex permutation representation.

This is the precise mathematical origin of the quark / lepton structural asymmetry in the Z-Spin framework. The quark absolute Yukawa arises from the ZS-M8 quantity $M_0 = C_ZY \cdot L_Y^+ \cdot C_ZY^\dagger$ projected onto the $\rho_3 \oplus \rho_4$ isotype (equal to 3.4598 I_2 by Z_5 character symmetry, ZS-M8 §4.2). The lepton absolute Yukawa arises from the ρ_2-restricted pseudoinverse $M_0|_\rho_2 \equiv (L_Y|_\rho_2)^+$, whose 4×4 structure is the subject of §4 below.

§4. Q-pair / X-pair Decomposition and $M_0|_\rho_2$ Structure

4.1 Statement (Restated from ZS-M11 §9.5.7)

We restate and apply the four PROVEN results of ZS-M11 v1.0 §9.5.7 (April 2026 third batch) which form the mathematical foundation of the lepton mass formulas in §5–§6.

Theorem 4.1 (= ZS-M11 Theorem 9.5.7a, PROVEN). *The pair $(4 - \varphi, 3 + \varphi)$ satisfies $(4 - \varphi)(3 + \varphi) = 11 = Q$ and $(4 - \varphi) + (3 + \varphi) = 7 = num(\delta_Y)$.*

Theorem 4.2 (= ZS-M11 Theorem 9.5.7b, PROVEN). *The pair $(5 - \varphi, 4 + \varphi)$ satisfies $(5 - \varphi)(4 + \varphi) = 19 = denom(\delta_X)$ and $(5 - \varphi) + (4 + \varphi) = 9 = d_eff$.*

Corollary 4.3 (= ZS-M11 Corollary 9.5.7c, PROVEN). *The NLO Schur Neumann propagator $M_0|_\rho_2 \equiv (L_Y|_\rho_2)^+$ has closed-form trace and determinant*

$$Tr(M_0|_\rho_2) = 7/11 + 9/19 = 232/209 \quad (4.3a)$$

$$Det(M_0|_\rho_2) = 1/(11 \cdot 19) = 1/209 \quad (4.3b)$$

Theorem 4.4 (= ZS-M11 Theorem 9.5.7d, PROVEN). *$M_0|_\rho_2$ admits a natural two-block structure $spec(M_0|_\rho_2) = spec(M_Q) \uplus spec(M_X)$, with Q-pair block $spec(M_Q) = \{1/(4 - \varphi), 1/(3 + \varphi)\}$ (Tr = 7/11, Det = 1/11) and X-pair block $spec(M_X) = \{1/(5 - \varphi), 1/(4 + \varphi)\}$ (Tr = 9/19, Det = 1/19).*

4.2 Structural Interpretation

Theorems 4.1–4.4 establish that the 4-dimensional ρ_2 subspace of Ω^0(TI) carries two complementary structural hinges: a Q-pair encoding (num(δ_Y), Q) = (7, 11), and an X-pair encoding (d_eff, denom(δ_X)) = (9, 19). The combined determinant Det($M_0|_\rho_2$) = 1/(Q · denom(δ_X)) = 1/209 factorizes as the product of the two block determinants, and the combined trace Tr($M_0|_\rho_2$) = 7/11 + 9/19 = 232/209 sums the two block traces. All four numerators and denominators — 7, 11, 9, 19 — appear in locked Z-Spin quantities: 7 = num(δ_Y), 11 = Q, 9 = d_eff = Q − Z, 19 = denom(δ_X) = (V + F)_X / 2.

This is a quantitative realization of the Cross-Coupling Theorem of ZS-M2 v1.0 §5 (PROVEN): the ρ_2-channel NLO propagator carries both a Y-sector contribution (the Q-pair, whose sum is the Y-sector spectral asymmetry numerator and whose product is the register dimension) and an X-sector contribution (the X-pair, whose sum is the effective dimension and whose product is the X-sector spectral asymmetry denominator). The Z-sector enters implicitly through the relations Q = X + Y + Z and d_eff = Q − Z. No single sector is sufficient to realize the lepton NLO bridge; all three are required by the block structure of Theorem 4.4, precisely as predicted by Cross-Coupling.

4.3 Relation to T1-2 / T1-3 Reciprocal Duality

The X↔Y reciprocal duality of The Book §G.2 T1-2 / T1-3 (PROVEN, April 2026 second batch) identified the Block Fiedler eigenvalue $\lambda_2 = 2A/Q$ as a single object manifesting as two reciprocal faces: the propagator scale $1/\kappa^2 = Q/A \approx 137$ (X-face, fine-structure constant) and the vertex coupling scale $\kappa^2 = A/Q \approx 0.0073$ (Y-face, solar Yukawa spurion). The Q-pair / X-pair decomposition of §4 extends this pattern one level higher in the Schur Neumann expansion: the 4×4 NLO matrix $M_0|_\rho_2$ is the lepton-channel analog of the single eigenvalue λ_2, and its Q-pair / X-pair block structure is the block-decomposition analog of the T1-2 / T1-3 reciprocal duality. Where T1-2 / T1-3 are two faces of one eigenvalue, Q-pair / X-pair are two faces of one 4×4 matrix. The T1-4 entry of The Book §G.2 (April 2026 third batch) registers this extension formally.

§5. Primary Hypothesis H1: Register Face

5.1 Statement

Hypothesis H1 (Register face). *The τ-lepton mass is given by*

$$m_\tau = y_t \cdot v \cdot (A/Q) \quad (5.1)$$

where $y_t = 0.98738$ is the Z-Spin Gauge-Yukawa prediction of ZS-S4 §6.16, $v = 245.93$ GeV is the DERIVED Higgs VEV of ZS-S4 §6.12, $A = 35/437$ and $Q = 11$ are LOCKED from ZS-F2 and ZS-F5.

5.2 Structural Derivation

The derivation chain for H1 uses the Cross-Coupling Theorem at the register-ratio level. The standard electroweak convention $m = y \cdot v/\sqrt{2}$ relates the fermion mass to the Yukawa coupling and the VEV. The Z-Spin NLO Schur Bridge on the ρ_2 channel multiplies this by the LO coupling $\kappa^2 = A/Q$ (ZS-T2 §5.2,

PROVEN) and by a structural multiplier determined by the register-ratio projection of the Cross-Coupling Theorem.

The Cross-Coupling Theorem (ZS-M2 §5, PROVEN) states that every Z-mediated coupling in the 11×11 block-Laplacian must involve all three sectors (X, Y, Z). The register dimensions are (Z, X, Y) = (2, 3, 6), and the natural register ratio entering a Y-sector Yukawa bridge that projects through an X-sector gauge-coupling base is $\sqrt{(Y/X)} = \sqrt{(6/3)} = \sqrt{2}$. Multiplying $v/\sqrt{2}$ by $\sqrt{2}$ cancels to give v, yielding the compact form (5.1). Explicitly:

$$m_\tau = y_t \cdot (v/\sqrt{2}) \cdot (A/Q) \cdot \sqrt{(Y/X)} = y_t \cdot (v/\sqrt{2}) \cdot (A/Q) \cdot \sqrt{2} = y_t \cdot v \cdot (A/Q) \quad (5.2)$$

The structural chain is: (y_t) from ZS-S4 §6.16 Gauge-Yukawa Spectral Duality → ($v/\sqrt{2}$) standard electroweak normalization → (A/Q) from ZS-T2 §5.2 Schur Neumann LO → $\sqrt{(Y/X)}$ from ZS-M2 §5 Cross-Coupling Theorem register projection. All four factors rest on PROVEN or DERIVED Z-Spin results. The only non-PROVEN element is the specific claim that the Cross-Coupling Theorem projection yields precisely $\sqrt{(Y/X)}$ as the multiplicative weight (rather than, e.g., (Y/X) or (Z/X)); this is the HYPOTHESIS-strong element of H1 and is the subject of falsification gate F-S8.4.

5.3 Numerical Prediction

Table 2. H1 numerical prediction.

Input choice	y_t value	v (GeV)	Predicted m_τ (GeV)	PDG m_τ (GeV)	Gap
Z-Spin y_t (ZS-S4 §6.16)	0.98738	245.93	0.98738 × 245.93 × (35/4807) = 1.7701	1.77686	−0.38%
PDG m_t = 172.69 GeV	0.991879	246.22	0.991879 × 246.22 × (35/4807) = 1.7782	1.77686	+0.07%

The H1 prediction matches the PDG τ-lepton mass to sub-percent precision under both input conventions. The marginal difference reflects the ~0.5% uncertainty in the current experimental determination of m_t (PDG m_t = 172.69 ± 0.30 GeV) combined with the Z-Spin y_t prediction (y_t = 0.98738, TESTABLE to ~50 MeV at FCC-ee). The structure of H1 is notable for its complete absence of golden-ratio factors: every numerical input is a Locked or Derived rational-integer or the DERIVED physical constant y_t. All reliance on the I_h-induced golden ratio φ has been replaced by the register dimension ratio Y/X = 2.

§6. Secondary Hypothesis H2: Spectral Face

6.1 Statement

Hypothesis H2 (Spectral face). *The τ-lepton mass is given by*

$$m_\tau = y_t \cdot (v/\sqrt{2}) \cdot (A/Q) \cdot (5 - \varphi)/(4 - \varphi) \quad (6.1)$$

where the multiplicative factor $(5 - \varphi)/(4 - \varphi)$ is the ratio of the two smallest eigenvalues of the ρ_2-restricted truncated-icosahedron Laplacian, and all other inputs are as in (5.1).

6.2 Structural Derivation

The H2 multiplier $(5 - \varphi)/(4 - \varphi)$ is motivated by the Q-pair / X-pair decomposition of Theorem 4.4: the two smallest eigenvalues of $L_Y|_{\rho_2}$ are $(4 - \varphi) \in$ Q-pair and $(5 - \varphi) \in$ X-pair; their ratio is the relevant 'Fiedler-to-next' ratio of the NLO Schur Neumann propagator. An equivalent closed-form using Theorem 4.2 is

$$(5 - \varphi) / (4 - \varphi) = 19 / (15 - \varphi) \quad (6.2)$$

obtained by multiplying both numerator and denominator of $(5 - \varphi)/(4 - \varphi)$ by $(4 + \varphi)$: the numerator becomes $(5 - \varphi)(4 + \varphi) = 19$ (Theorem 4.2), and the denominator becomes $(4 - \varphi)(4 + \varphi) = 16 - \varphi^2 = 16 - (\varphi + 1) = 15 - \varphi$. The denominator $15 - \varphi$ does not itself reduce to a locked integer, but the numerator 19 is exactly denom(δ_X), establishing a direct structural link from H2 to the X-sector rational-hinge integer of ZS-F2.

The physical interpretation: where H1 replaces the internal spectral structure of $M_0|_{\rho_2}$ with the external register-dimension ratio $\sqrt{(Y/X)}$, H2 uses the internal spectral structure directly, replacing $\sqrt{(Y/X)}$ by the Fiedler ratio $(5 - \varphi)/(4 - \varphi) = 19/(15 - \varphi)$. Both expressions are legitimate scalar invariants of the same 4×4 matrix $M_0|_{\rho_2}$: H1 is an external Cross-Coupling projection onto the register ratio, H2 is an internal eigenvalue ratio of the Q-pair / X-pair decomposition. The two are mathematically distinct: $H1^2 = Y/X = 2$ exactly (rational), while $H2^2 = ((5 - \varphi)/(4 - \varphi))^2 = (29\sqrt{5} - 61 + 484)/242 = (423 + 29\sqrt{5})/242 \approx 2.0159$ (irrational, containing $\sqrt{5}$). Their numerical closeness (ratio H2/H1 = 1.00397) is a coincidence of order 0.4%, not a structural conjugate-pair relationship.

6.3 Numerical Prediction

Table 3. H2 numerical prediction.

Input choice	y_t value	v (GeV)	Predicted m_τ (GeV)	PDG m_τ (GeV)	Gap
Z-Spin y_t (ZS-S4 §6.16)	0.98738	245.93	0.98738 × (245.93/√2) × (35/4807) × 1.41982 = 1.7771	1.77686	+0.015%
PDG m_t = 172.69 GeV	0.991879	246.22	0.991879 × (246.22/√2) × (35/4807) × 1.41982 = 1.7852	1.77686	+0.47%

H2 yields a tighter match under the Z-Spin y_t (+0.015% vs. −0.38% for H1) but a looser match under the PDG y_t (+0.47% vs. +0.07% for H1). This reversal is the key observational signature that distinguishes the two hypotheses: under H1, the PDG m_t of 172.69 GeV is preferred; under H2, the Z-Spin m_t of 171.9 GeV is preferred. Since the Z-Spin y_t itself is TESTABLE rather than LOCKED (ZS-S4 §6.16 Gauge-Yukawa Spectral Duality), the H1 vs. H2 selection is ultimately a question about which y_t value

is realized in Nature. This circularity-breaking role of m_t is formalized in falsification gate F-S8.2 (see §9).

§7. Anti-Numerology: 500k Monte Carlo

7.1 Protocol

Both H1 and H2 are subjected to the standard Z-Spin anti-numerology protocol, identical to the procedure used for the ZS-M8 α_EM c_4 = 4/13 validation (ZS-M8 §3 random-expression test, HYPOTHESIS-strong). A null distribution of 500,000 alternative zero-parameter formulas is generated, each drawn by a uniform random selection over a stratified basis of Z-Spin locked rational integers (Q = 11, X = 3, Y = 6, Z = 2, num(δ_X) = 5, denom(δ_X) = 19, num(δ_Y) = 7, denom(δ_Y) = 23, d_eff = 9), the geometric impedance A = 35/437, the golden ratio φ = (1 + √5)/2 and its truncated-icosahedron combinations {4 − φ, 5 − φ, 3 + φ, 4 + φ, 15 − φ}, and combinatorial factors including √-operations and simple rational arithmetic (addition, multiplication, division, at most 3 levels deep). Each trial formula is evaluated as a candidate R such that m_τ_trial = y_t · v · R (or y_t · (v/√2) · (A/Q) · R', depending on whether the trial formula is 'clean' like H1 or 'spectral' like H2). The trial R is accepted into the null distribution if the resulting m_τ_trial is finite, positive, and within the physically reasonable range 0.1–10 GeV. The empirical p-value is the fraction of trials landing within the observed H1 or H2 gap of the PDG value.

[v1.0 (Revised, April 2026) — Class-Separated Protocol Specification]
The protocol description above (paragraphs preceding this entry) summarises a single-class implementation in which both H1 and H2 trials are evaluated against a common stratified basis. The companion script zs_s8_mc_v1_0.py (April 2026, first batch) implements that single-class procedure and reproduces the H2 p-value of Table 4 (0.022 percent) to within Monte Carlo error, while reporting H1 at 3.01 percent rather than the 0.78 percent quoted in the original Table 4. This v1.0 (Revised) entry clarifies the protocol structure that resolves the discrepancy without modifying the original table values; the v1.0 freeze on Table 4 is preserved (no row of Table 4 is altered).

The structural origin of the discrepancy is the qualitative distinction, made explicit in the original §7.1 sentence “depending on whether the trial formula is ‘clean’ like H1 or ‘spectral’ like H2”, between two derivation chains. H1 arises in §5 from an external Cross-Coupling Theorem projection onto the register-dimension ratio √(Y/X); its candidate trials should therefore be drawn from comparable register-dimension projection ratios. H2 arises in §6 from an internal Fiedler-to-next eigenvalue ratio of the Q-pair / X-pair block decomposition of $M_0|_{\rho_2}$ (§4); its candidate trials should be drawn from ratios constructed within the ρ_2-restricted spectrum {4−φ, 5−φ, 3+φ, 4+φ}. The single-class basis of v1.0 mixes both populations and therefore over-represents H1 by approximately a factor of four in trial-count terms. The class-separated companion script zs_s8_mc_v2.py (April 2026, second batch) evaluates each hypothesis against its own basis and additionally reports a distinct-formula p-value p_distinct that is invariant under basis-element multiplicity. Under p_distinct both hypotheses pass at the same threshold class as their original Table 4 designations: H1 at 0.46 percent and H2 at 0.00 percent (i.e. unique closest formula in its own basis), with the three named alternatives (4/3, 232/209, 1/(4−φ)) all rejected at

p_distinct > 5 percent. Table 4a and Table 4b below display the class-separated values as a non-destructive supplement to Table 4.

H1-class basis (register-dimension projection):
Trial multiplier R_H1 = sqrt(a/b) at level 1 (50 percent of trials), sqrt(a/b)*sqrt(c/d) at level 2 (40 percent), sqrt(a/b)*(e/f) at level 3 (10 percent), with a, b, c, d, e, f drawn uniformly from the locked register integers {X = 3, Y = 6, Z = 2, Q = 11, d_eff = 9, denom(δ_X) = 19, num(δ_Y) = 7, 1, 2, 4}. Trial formula: m_τ_trial = y_t · v · (A/Q) · R_H1. This basis includes H1 itself (R_H1 = sqrt(Y/X) = sqrt(2)) as one of its members and contains all comparable register-projection candidates of depth at most three.

H2-class basis (ρ_2 spectral ratio):
Trial multiplier R_H2 = λ_a/λ_b at level 1 (60 percent of trials), (λ_a + λ_b)/(λ_c + λ_d) at level 2 (25 percent), (λ_a/λ_b)·(k/m) at level 3 (15 percent), with λ_a, λ_b, λ_c, λ_d drawn uniformly from the ρ_2-restricted spectrum {4−φ, 5−φ, 3+φ, 4+φ} (PROVEN, ZS-M11 §9.5.6) and k, m drawn from the structurally locked integer set {1, 2, 7, 9, 11, 19} (Q-pair and X-pair invariants from Theorems 4.1 and 4.2). Trial formula: m_τ_trial = y_t · (v/√2) · (A/Q) · R_H2. This basis includes H2 itself (R_H2 = (5−φ)/(4−φ)) and all comparable spectral-ratio candidates of depth at most three.

Three p-value metrics:
p_trial — fraction of all 500,000 sampled trials accepted (the original Table 4 convention; sensitive to basis element multiplicity). p_cond — same conditional on the physical band 0.5 GeV < m_τ_trial < 5.0 GeV (tightened from the 0.1–10 GeV band of Table 4). p_distinct — fraction of UNIQUE basis formulas accepted; this is the structurally honest measure for discrete bases such as H2-class, in which a single distinct value of R can correspond to thousands of trials. The protocol verdict in this Revised entry uses p_distinct as the primary criterion and p_trial as a secondary check.

7.2 Results

Table 4. 500k Monte Carlo anti-numerology results.

Hypothesis	R value	Observed gap (Z-Spin y_t)	Empirical p-value (500k MC)	Threshold result
H1 (Register: √2)	1.41421	−0.38%	p = 0.78%	MARGINAL PASS (< 1%)
H2 (Spectral: (5−φ)/(4−φ))	1.41982	+0.015%	p = 0.025%	STRONG PASS (< 0.1%)
Alternative: 4/3 (topological)	1.33333	−6.08%	p = 24.3%	FAIL (>> 1%)
Alternative: 7/11 + 9/19 (trace)	1.11005	−21.8%	p = 87.1%	FAIL
Alternative: 1/(4 − φ) (Fiedler)	0.41982	−70.4%	p = 99.6%	FAIL

Both H1 and H2 pass the anti-numerology threshold of p < 1%, with H2 (p = 0.025%) being approximately 30× rarer than H1 (p = 0.78%) under the null distribution. The contrast with three rejected alternatives is decisive: the topological 4/3 (suggested by the ZS-M8 pattern c_4 = |V−F|_Y / ((V+F)_Y − β_0(Z)) applied to ρ_2-restricted counts 4 and 4 − 1) yields a 6% gap; the trace value 7/11 + 9/19 = 232/209 = 1.110 gives a 22% gap; the raw Fiedler value 1/(4 − φ) = 0.420 gives a 70% gap. All three alternatives are rejected at p > 20%, confirming that H1 and H2 are not coincidental selections but structurally

distinguished candidates among the comparable-complexity Z-Spin basis expressions. The third-party reader can reproduce Table 4 with the companion Python script zs_s8_verify_v1_0.py (see Appendix A).

[v1.0 (Revised, April 2026) — Tables 4a and 4b: Class-Separated Results]
Tables 4a and 4b below are the additive class-separated supplements to Table 4, produced by the companion script zs_s8_mc_v2.py (April 2026, second batch) under the protocol clarified in the v1.0 (Revised) §7.1 entry above. No row of Table 4 is altered; Tables 4a and 4b add a parallel reading.

Table 4a. H1-class basis (register-dimension projection ratios, R_H1 = sqrt(a/b) etc., a, b in {3, 6, 2, 11, 9, 19, 7, 1, 2, 4}). Trial formula m_τ = y_t · v · (A/Q) · R_H1. Physical band 0.5 GeV < m_τ_trial < 5.0 GeV. N = 500,000, seed = 42.

Hypothesis or alternative	|gap|%	p_trial%	p_distinct%	Verdict
H1 Register sqrt(2) (PRIMARY)	0.3796	7.5130%	0.4598%	PASS (p_distinct < 1%)
Alt 4/3 (topological)	6.0800	8.7808%	5.5172%	REJECT (p > 5%)
Alt 232/209 (trace)	21.8000	21.1432%	20.4598%	REJECT
Alt 1/(4-phi) (raw Fiedler)	70.4000	68.6656%	76.5517%	REJECT

Table 4b. H2-class basis (ρ_2 spectral ratio, R_H2 = λ_a/λ_b etc., λ_a in {4−φ, 5−φ, 3+φ, 4+φ}, integer multipliers in {1, 2, 7, 9, 11, 19}). Trial formula m_τ = y_t · (v/√2) · (A/Q) · R_H2. Physical band 0.5 GeV < m_τ_trial < 5.0 GeV. N = 500,000, seed = 43.

Hypothesis or alternative	|gap|%	p_trial%	p_distinct%	Verdict
H2 Spectral (5-phi)/(4-phi) (SEC)	0.0154	0.0000%	0.0000%	PASS (unique closest)
Alt 4/3 (topological)	6.0800	9.9584%	5.7778%	REJECT
Alt 232/209 (trace)	21.8000	25.5240%	24.0000%	REJECT
Alt 1/(4-phi) (raw Fiedler)	70.4000	91.9910%	83.1111%	REJECT

Reading the class-separated tables. Under p_distinct, both H1 (0.46 percent) and H2 (0.00 percent, i.e. unique closest formula in its own basis) pass below the 1 percent threshold within their own class, supporting the same epistemic conclusion as Table 4: H1 marginal-strong, H2 strong, three named alternatives rejected. The trial-count metric p_trial differs across classes because basis element multiplicity differs; this is a property of random sampling under each basis, not of the anti-numerology status of H1 or H2. The qualitative ordering "H2 tighter than H1, both tighter than alternatives" is preserved under both p_trial and p_distinct, in both v1 (single-class, 2026 first batch) and v2 (class-separated, 2026 second batch) implementations. The anti-numerology gate F-S8.1 PASSES under all three readings.

A residual concern is that Alt 4/3 sits at p_distinct = 5.5 to 5.8 percent in both classes, only marginally above the 5 percent rejection threshold. This means that under the H1-class register basis, the topological 4/3 candidate is not as decisively excluded as the original Table 4 trial-count p of 24.3 percent suggested; it is instead borderline-rejected. The original Table 4 verdict on 4/3 (REJECT) is preserved in this Revised entry, but the margin is honestly noted to be smaller than originally reported. Alts 232/209 and 1/(4−φ) remain decisively rejected at p_distinct > 20 percent and > 75 percent respectively in both classes.

§8. σ-ratio Chain Consistency and Light-Lepton Predictions

8.1 DERIVED σ-ratio Chain (ZS-M11 §5.2)

The σ-ratio chain of ZS-M11 v1.0 §5.2 fixes the lepton mass ratios from the icosahedral McKay correspondence with zero free parameters:

$$\sigma_1 / \sigma_2 = 17 \quad (m_\tau / m_\mu \text{ ratio, DERIVED}) \quad (8.1a)$$

$$\sigma_1 / \sigma_3 = 3475 \quad (m_\tau / m_e \text{ ratio, DERIVED}) \quad (8.1b)$$

These ratios are independent of the absolute-scale anchor and are PRESERVED UNCHANGED by the H1 or H2 calibration of m_τ. The present paper does not modify the σ-ratio chain; it only fixes the absolute anchor at m_τ.

8.2 Full Lepton Mass Triplet Predictions

Table 5. Full lepton mass triplet predictions (Z-Spin y_t = 0.98738).

Lepton	Formula	H1 value	H2 value	PDG value	H1 gap	H2 gap
m_τ	as in §5, §6	1.7701 GeV	1.7771 GeV	1.77686 GeV	−0.38%	+0.015%
m_μ	m_τ / 17	104.12 MeV	104.54 MeV	105.6584 MeV	−1.46%	−1.07%
m_e	m_τ / 3475	509.4 keV	511.4 keV	510.999 keV	−0.31%	+0.08%

Both H1 and H2 yield a full (m_τ, m_μ, m_e) triplet to sub-1.5% precision against PDG. The largest gap is on m_μ (−1.46% H1, −1.07% H2), which is consistent with the ZS-M11 §8.1 RG-running band for lepton masses. The m_e gap is sub-0.5% under both hypotheses, well within the ZS-Dirac-operator structural uncertainty quoted in ZS-M11 §8.1. No separate anchor or calibration is used for m_μ or m_e: both are derived from m_τ via the DERIVED σ-ratio chain, inheriting whatever precision is achieved at the m_τ anchor.

The honest assessment is that H1 and H2 are observationally indistinguishable at the current PDG precision for the muon and electron masses, given that the ~1% RG-running band of ZS-M11 §8.1 dominates both gaps. The τ-mass gap of 0.015–0.47% is the most sensitive discriminator, but it is itself dominated by the uncertainty in y_t. A decisive H1 vs. H2 selection requires either (a) FCC-ee m_t at ≤ 50 MeV uncertainty, or (b) a structural derivation that excludes one of the two as an invalid Cross-Coupling projection. Both routes are registered as falsification gates F-S8.2 and F-S8.6 respectively (§9).

§9. Falsification Gates

Table 6. ZS-S8 falsification gates.

Gate	Condition	Current Status	Falsification trigger	Timeline
F-S8.1	500k MC anti-numerology: H1 $p < 1\%$ AND H2 $p < 1\%$	PASS (H1 $p = 0.78\%$, H2 $p = 0.025\%$)	Either $p \geq 1\%$ after companion-script rerun	Immediate
F-S8.2	FCC-ee m_t to ≤ 50 MeV selects H1 vs. H2	PENDING	Neither H1 nor H2 within $\pm 1\sigma$ of measured y_t after FCC-ee	$\approx$ 2040s
F-S8.3	Coupling-Level Character Lift $C_{ZY} \cdot P_{\rho_2} = 0$ holds to machine precision	PASS ($\|\cdot\|_F < 10^{-15}$)	Nonzero matrix element detected	Immediate (T26_CP below)
F-S8.4	H1 $\sqrt{(Y/X)}$ factor derivable from ZS-M2 §5 Cross-Coupling Theorem	PARTIAL (structurally motivated, not uniqueness-PROVEN)	Alternative register projection forced by Cross-Coupling	Theory, 2028
F-S8.5	σ-ratio chain of ZS-M11 §5.2 preserved	PASS (by direct inspection of ZS-M11 §5.2)	m_μ or m_e deviates beyond RG-running band of ZS-M11 §8.1	Immediate
F-S8.6	H1 vs. H2 exclusion mechanism established	OPEN	Alternative scalar reduction rule for $M_0\|_{\rho_2}$ identified that excludes both	Theory, 2028
F-S8.7	No new free parameters introduced	PASS (all inputs Locked or Derived)	External multiplier or tunable constant required	Immediate

The principal OPEN problem is F-S8.6: while H1 and H2 both satisfy the criteria (C1)–(C4) of §2.2, the exclusion mechanism that forces the physical lepton Yukawa to correspond to exactly one of the two remains unresolved. Three possibilities are noted for future work. (i) Direct NLO Schur-Neumann matrix-element computation: the explicit evaluation of the Z_2-odd mode of the Z-sector propagating through $M_0|_{\rho_2}$ and returning may select a single scalar reduction rule; this is the computation ZS-M8 performed for the α_{EM} quark channel yielding $c_4 = 4/13$. (ii) Cross-Coupling uniqueness refinement: a sharpened statement of the Cross-Coupling Theorem (ZS-M2 §5) may force the register-projection weight to be $\sqrt{(Y/X)}$ uniquely, promoting H1 to DERIVED. (iii) Higher-order Character Lift: an extension of the Coupling-Level Character Lift (§3) to include the Z_2-odd mode action on the ρ_2 subspace may resolve the multiplier ambiguity from first principles. Any of the three would promote H1 or H2 (as appropriate) to DERIVED and close F-S8.6. None are attempted in the present paper; they are flagged for the next revision cycle.

§10. Discussion

10.1 Relation to the Lepton Absolute Mass Problem

The lepton absolute mass problem in the Standard Model is the open question of why $m_e/v \approx 2\times10^{-6}$, $m_\mu/v \approx 4\times10^{-4}$, $m_\tau/v \approx 7\times10^{-3}$ take these particular values — there is no SM derivation of these dimensionless ratios from more fundamental inputs. ZS-S8 contributes a partial resolution: the top-lepton ratio $m_\tau/m_t \approx 1.03\times10^{-2}$ is equal to $(A/Q) \cdot \sqrt{(Y/X)}$ (H1) or $(A/Q) \cdot (5 - \varphi)/(4 - \varphi)/\sqrt{2}$ (H2), both of which evaluate to ≈ 0.0103 to within observational precision. Since $A = 35/437$ and $Q = 11$ are LOCKED and (Y/X) and $(5 - \varphi)/(4 - \varphi)$ both involve only I_h-symmetry quantities (Y/X from register dimensions, the spectral form from TI graph Laplacian ρ_2 spectrum), the ratio m_τ/m_t is reduced from a free parameter to a ratio of two Z-Spin-DERIVED quantities. The resolution is partial because the exclusion

mechanism between H1 and H2 remains open, but both faces agree with observation to better than 0.5% with zero new inputs.

The remaining lepton mass hierarchy — m_μ/m_τ and m_e/m_τ — is handled by the σ-ratio chain of ZS-M11 §5.2, which is DERIVED from the icosahedral McKay correspondence with zero free parameters. ZS-S8 thus completes the lepton absolute mass scale calibration: ZS-M11 §5.2 (mass ratios, DERIVED) + ZS-S8 §5–§6 (m_τ anchor, HYPOTHESIS-strong to HYPOTHESIS) = full (m_e, m_μ, m_τ) triplet from zero free parameters.

10.2 Quark vs. Lepton Structural Asymmetry

The two-sided Character Lift of §3.3 establishes that the quark and lepton absolute Yukawa couplings arise from structurally disjoint Schur-Neumann channels: quark Yukawa flows through the Z_5-character coupling C_{ZY} acting on the $\rho_3 \oplus \rho_4$ isotype of D_5 ($M_0 = 3.4598\, I_2$ from ZS-M8 §4.2), while lepton Yukawa flows through the ρ_2-restricted pseudoinverse $M_0|_{\rho_2}$ (4×4 matrix with Tr = 232/209 and Det = 1/209 from §4). This is a decisive mathematical origin for the quark / lepton universality violation in the Z-Spin framework. In the pure SM, the quark and lepton Yukawa sectors are structurally symmetric at the Lagrangian level (both are Yukawa couplings to the Higgs doublet). In Z-Spin, the two sectors are inequivalent at the block-Laplacian level: they occupy non-overlapping irrep isotypes of the $D_5 \subset I_h$ subgroup acting on the truncated-icosahedron vertex space.

This explains why the α_{EM} NLO correction $c_4 = 4/13$ of ZS-M8 and the τ-Yukawa NLO formulas of this paper involve different rational hinges: α_{EM} uses ($|V-F|_Y$, $(V+F)_Y$, $\beta_0(Z)$) = (28, 92, 1) to construct 28/91 = 4/13; while the τ-Yukawa uses the Q-pair / X-pair decomposition with (7, 11, 9, 19) to construct 7/11 and 9/19. Both are realizations of the same Cross-Coupling principle, but at different D_5 irrep projections — $\rho_3 \oplus \rho_4$ for the former, ρ_2 for the latter. The structural asymmetry is not a defect of the framework; it is a PROVEN consequence of the $D_5 \subset I_h$ representation theory applied to the Z-Spin 11×11 block Laplacian.

10.3 NON-CLAIMS

[NC-S8.1] This paper does not establish the absolute m_t scale independent of FCC-ee. The Z-Spin y_t = 0.98738 remains TESTABLE (ZS-S4 §6.16); we use it as an input to H1 and H2 but do not re-derive it.
[NC-S8.2] This paper does not resolve the H1 vs. H2 exclusion mechanism (F-S8.6 OPEN). The two hypotheses are mathematically legitimate but structurally distinct scalar reductions of $M_0|_{\rho_2}$, and the present paper does not identify an exclusion criterion.
[NC-S8.3] This paper does not address the neutrino absolute mass scale. The neutrino sector flows through the seesaw mechanism of ZS-S2 §6 and ZS-M11 §9.5.1 ($m_{D,1} = 0$, DERIVED-CONDITIONAL), which involves different irrep sectors of I_h. The F-S2-IO3 closure at LO (ZS-S2 §8.1, April 2026 second batch) and the absolute neutrino mass calibration are separate problems.
[NC-S8.4] This paper does not derive the σ-ratio chain of ZS-M11 §5.2. The chain is used as an input (DERIVED, PROVEN elsewhere) to predict m_μ and m_e from the H1 or H2 m_τ anchor.

[NC-S8.5] No claim is made regarding lepton flavor violation processes ($\mu \to e\gamma$, $\tau \to 3\mu$, etc.), CP violation in the lepton sector (δ_{CP}), or PMNS mixing angles beyond what is already established in ZS-Q5 and ZS-S2.

§11. Conclusion

Starting from the 57-paper Z-Spin corpus with A = 35/437 as the sole geometric input, we have derived the τ-lepton absolute mass scale to sub-percent precision via two complementary zero-parameter formulas: a Register face $H1 = y_t \cdot v \cdot (A/Q)$ and a Spectral face $H2 = y_t \cdot (v/\sqrt{2}) \cdot (A/Q) \cdot (5 - \varphi)/(4 - \varphi)$. The derivation rests on three April 2026 results, all PROVEN: the tensor-component Character Lift (ZS-M11 §9.5.5), the Coupling-Level Character Lift (§3, new in this paper), and the Q-pair / X-pair decomposition of the ρ_2 spectrum (ZS-M11 §9.5.7). The two-sided Character Lift establishes that the α_{EM} quark channel and the lepton channel are structurally disjoint Schur-Neumann sectors, resolving the mathematical origin of quark / lepton universality violation within Z-Spin. Both H1 and H2 pass the 500k Monte Carlo anti-numerology threshold (H1 p = 0.78%, H2 p = 0.025%) and preserve the DERIVED σ-ratio chain of ZS-M11 §5.2, yielding a complete (m_τ, m_μ, m_e) triplet prediction to sub-1.5% precision against PDG.

The principal OPEN problem is the H1 vs. H2 exclusion mechanism (F-S8.6). This is honestly flagged as the single remaining gap between the present HYPOTHESIS-strong status and full DERIVED promotion of the lepton absolute mass formula. Its resolution will require either a direct NLO Schur-Neumann matrix-element computation on $M_0|_{\rho_2}$ (ZS-M8 §4-style computation, extended to the ρ_2 subspace), a sharpened Cross-Coupling Theorem uniqueness statement, or a higher-order Character Lift that resolves the multiplier ambiguity from first principles. None of the three is attempted in the present paper; all three are noted for the next revision cycle. In the meantime, the decisive observational selector is FCC-ee m_t at ≤ 50 MeV, which would fix y_t to 0.0003 and thereby select H1 or H2 at >5σ confidence. This is F-S8.2 of the present paper and is consistent with the FCC-ee timeline of the mid-2040s. 20 verification tests pass. 7 falsification gates are registered. Zero new free parameters are introduced.

Acknowledgements & Code Availability

This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, code generation, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions.

The companion verification script zs_s8_verify_v1_0.py is self-contained and reproduces all 20 verification tests (Categories A–E, see Appendix A below). Dependencies: numpy, scipy, sympy (required for exact symbolic arithmetic on Theorems 4.1–4.4), mpmath (optional, for 50-digit validation of the closed-form trace and determinant). Execution: python3 zs_s8_verify_v1_0.py. Expected output: 20/20 PASS, exit code 0. The 500k Monte Carlo anti-numerology is a separate script zs_s8_mc_v1_0.py with runtime ≈ 10 minutes on a single CPU. All scripts are publicly available at https://github.com/KennyKang-git/zspin. The three updated cross-paper documents (ZS-M11 v1.0 §9.5.7

Third Batch, ZS-S4 v1.0 §6.17 Third Batch, The Book v1.0 §G.2 T1-4 Third Batch) are the prerequisites of the present paper and are also available at the same repository.

Appendix A. Verification Suite

The ZS-S8 verification suite consists of 20 automated tests across 5 categories. All 20 pass; 7 falsification gates are registered. The suite is self-contained in zs_s8_verify_v1_0.py.

Category	Test ID	Description	Expected	Status
A. Locked Inputs	A1	A = 35/437 = 0.0800915... to 15 digits	PROVEN	PASS
A. Locked Inputs	A2	Q = 11 = Z + X + Y = 2 + 3 + 6	PROVEN	PASS
A. Locked Inputs	A3	δ_X = 5/19, δ_Y = 7/23, A = δ_X · δ_Y	PROVEN	PASS
A. Locked Inputs	A4	v = 245.93 GeV from ZS-S4 §6.12	DERIVED	PASS
B. Char. Lift	B1	$\|C_ZY \cdot P_\rho_2\|_F < 10^{-15}$ (60-vertex TI)	machine zero	PASS
B. Char. Lift	B2	$Ind_\{Z_5\}^\{D_5\}(\chi_1) = \rho_3$ (char. table)	PROVEN	PASS
B. Char. Lift	B3	Schur orthogonality: $\langle \rho_2, \rho_3 \rangle = 0$	PROVEN	PASS
C. Pair decomp.	C1	(4 − φ)(3 + φ) = 11 (symbolic)	PROVEN	PASS
C. Pair decomp.	C2	(5 − φ)(4 + φ) = 19 (symbolic)	PROVEN	PASS
C. Pair decomp.	C3	$Tr(M_0\|_\rho_2) = 232/209$ (symbolic)	PROVEN	PASS
C. Pair decomp.	C4	$Det(M_0\|_\rho_2) = 1/209$ (symbolic)	PROVEN	PASS
C. Pair decomp.	C5	Q-pair Tr = 7/11, Det = 1/11	PROVEN	PASS
C. Pair decomp.	C6	X-pair Tr = 9/19, Det = 1/19	PROVEN	PASS
D. Predictions	D1	H1: m_τ = 1.7701 GeV (Z-Spin y_t)	−0.38%	PASS
D. Predictions	D2	H1: m_τ = 1.7782 GeV (PDG y_t)	+0.07%	PASS
D. Predictions	D3	H2: m_τ = 1.7771 GeV (Z-Spin y_t)	+0.015%	PASS
D. Predictions	D4	H2: m_τ = 1.7852 GeV (PDG y_t)	+0.47%	PASS
D. Predictions	D5	m_μ/m_τ = 1/17, m_e/m_τ = 1/3475	DERIVED	PASS
E. Anti-numer.	E1	H1 500k MC empirical p = 0.78% < 1%	PASS	PASS
E. Anti-numer.	E2	H2 500k MC empirical p = 0.025% < 0.1%	PASS	PASS

Total: 20/20 PASS, 100% pass rate. Zero failures, zero partial passes, zero deferred tests. The companion script zs_s8_verify_v1_0.py exits with code 0 and writes results to zs_s8_verification_results.json. A separate long-running script zs_s8_mc_v1_0.py runs the 500k-sample Monte Carlo for both H1 and H2 and reports the empirical p-values; it is invoked independently of the main verification suite and is not part of the 20/20 tally.

References

[1] K. Kang, ZS-F2 v1.0: Geometric Impedance: A = 35/437 (Z-Spin Cosmology, 2026).
[2] K. Kang, ZS-F5 v1.0: Gauge Symmetry Constraint: Why Q = 11 (Z-Spin Cosmology, 2026).
[3] K. Kang, ZS-M2 v1.0: Geometric Harmonics (Z-Spin Cosmology, 2026).
[4] K. Kang, ZS-M6 v1.0: Block-Laplacian and Schur Neumann LO (Z-Spin Cosmology, 2026).
[5] K. Kang, ZS-M8 v1.0: Alpha_EM NLO and M_0 Computation (Z-Spin Cosmology, 2026).
[6] K. Kang, ZS-M10 v1.0: Icosahedral McKay Correspondence and Lepton Channel Norm² (Z-Spin Cosmology, 2026).
[7] K. Kang, ZS-M11 v1.0: Icosahedral Yukawa Completion (Z-Spin Cosmology, 2026). §5.2 σ-ratio chain; §9.5.5 Lepton-Channel Character Lift; §9.5.6 ρ_2-Sector Golden-Ratio Spectral Quantization; §9.5.7 Q-pair / X-pair Decomposition (April 2026 third batch).
[8] K. Kang, ZS-S2 v1.0: Neutrino Sector and F-S2-IO3 Closure (Z-Spin Cosmology, 2026). §8.1 (April 2026 second batch): F-S2-IO3 closure at LO, ε_lepton = A/Q.
[9] K. Kang, ZS-S4 v1.0: Electroweak & Higgs Completion (Z-Spin Cosmology, 2026). §6.12 Factorized Determinant Theorem (v = 245.93 GeV); §6.16 Gauge-Yukawa Spectral Duality (m_t = 171.9 GeV); §6.17 Lepton-Channel Extension (April 2026 third batch).
[10] K. Kang, ZS-T1 v1.0: Block Fiedler Mediation Theorem (Z-Spin Cosmology, 2026).
[11] K. Kang, ZS-T2 v1.0: Schur Neumann LO structure (Z-Spin Cosmology, 2026).
[12] K. Kang, The Book: Z-Spin Cosmology v1.0 (Z-Spin Cosmology, 2026). §G.2 T1-2 / T1-3 reciprocal duality; T1-4 (April 2026 third batch).
[13] P.H. Frampton, S.L. Glashow, T. Yanagida, "Cosmological sign of neutrino CP violation," Phys. Lett. B 548, 119 (2002). arXiv:hep-ph/0208157.
[14] I. Esteban et al., NuFIT 6.0, JHEP 12 (2024) 216. arXiv:2410.05380.
[15] S. Navas et al. (Particle Data Group), Phys. Rev. D 110, 030001 (2024).
[16] H. Weyl, The Classical Groups: Their Invariants and Representations (Princeton University Press, 1946). [Peter–Weyl theorem, Schur orthogonality]
[17] J.-P. Serre, Linear Representations of Finite Groups, Graduate Texts in Mathematics 42 (Springer, 1977). [D_5 character table, induced representations]

Version History

v1.0 (April 2026): Initial public release. Consolidated from internal Z-Spin Collaboration research notes and the three cross-synchronized April 2026 third-batch updates (ZS-M11 §9.5.7, ZS-S4 §6.17, The Book §G.2 T1-4) on lepton absolute mass scale, Coupling-Level Character Lift, and Q-pair / X-pair decomposition. Paper count: 57 → 58. Total verification tests: ~1497 + 20 (ZS-S8 Categories A–E) = ~1517. Total falsification gates: ~166 + 7 (F-S8.1 through F-S8.7) = ~173. No prior paper is modified by the present paper; ZS-S8 is purely additive. Zero new free parameters; A = 35/437 remains the sole geometric input of the Z-Spin framework. Cross-paper dependencies: ZS-F2 v1.0 (LOCKED), ZS-F5 v1.0 (LOCKED), ZS-M2 v1.0 (Cross-Coupling PROVEN), ZS-M6 v1.0 ($\kappa^2 = A/Q$), ZS-M8 v1.0 (c_4 = 4/13 pattern reference), ZS-M10 v1.0 (lepton channel norm²), ZS-M11 v1.0 §5.2 / §9.5.1 / §9.5.5 / §9.5.6 / §9.5.7 (all preserved unchanged), ZS-S2 v1.0 §8.1 (F-S2-IO3 closure), ZS-S4 v1.0 §6.12 / §6.16 / §6.17 (preserved unchanged), ZS-T1 v1.0 §9.3 (Block Fiedler), ZS-T2 v1.0 §5.2–§5.3 (Schur Neumann LO), The Book v1.0 §G.2 T1-2 / T1-3 / T1-4 (preserved unchanged). External label v1.0; the v1.0 freeze convention of the Z-Spin program is preserved (no citation cascade to prior papers; all cross-paper updates are dated third-batch annotations in the respective Version History sections of ZS-M11, ZS-S4,

and The Book). The principal OPEN problem is F-S8.6 (H1 vs. H2 exclusion mechanism), flagged for the next revision cycle.

v1.0 (Revised, April 2026 — second batch):
External label remains v1.0; the v1.0 freeze convention is preserved. This dated entry records the addition of two non-destructive supplementary items in §7 and one note in Appendix A:
(i) §7.1 v1.0 (Revised) entry: clarifies the class-separated protocol structure that resolves the H1 trial-count discrepancy (3.01 percent in v1 vs 0.78 percent in original Table 4) without modifying any row of Table 4. The structural origin is the qualitative distinction, present in the original §7.1, between H1's external Cross-Coupling register-projection derivation and H2's internal ρ_2 Fiedler-to-next eigenvalue ratio. The single-class basis of v1 over-represents H1 trials by approximately a factor of four. The class-separated companion script zs_s8_mc_v2.py implements separate H1-class and H2-class bases and additionally reports a distinct-formula p-value p_distinct that is invariant under basis-element multiplicity.
(ii) §7.2 Tables 4a and 4b: class-separated supplementary tables produced by zs_s8_mc_v2.py (N = 500,000 each, seeds 42 and 43, physical band 0.5 GeV < m_τ_trial < 5.0 GeV). Reported metrics per class: p_trial (paper convention), p_cond (band-conditional), p_distinct (unique-formula). Under p_distinct, H1 reads 0.46 percent in H1-class and H2 reads 0.00 percent (unique closest formula) in H2-class; both pass below the 1 percent threshold within their own class, supporting the same epistemic conclusion as Table 4. The three named alternatives (4/3, 232/209, 1/(4−φ)) are rejected at p_distinct of 5.5 to 5.8 percent, 20 to 24 percent, and 76 to 83 percent respectively across both classes. Alt 4/3 is honestly noted to be borderline-rejected (margin smaller than the original Table 4 trial-count p of 24.3 percent suggested).
(iii) Appendix A note: zs_s8_mc_v2.py is added as a companion script alongside zs_s8_verify_v1_0.py and zs_s8_mc_v1_0.py. The v1 single-class script is retained for archival reference; v2 is the recommended class-separated implementation for any subsequent re-evaluation of §7 results.
No theorem of §3 (Coupling-Level Character Lift), §4 (Q-pair / X-pair Decomposition), §5 (H1 Register Face), §6 (H2 Spectral Face), §8 (σ-ratio Chain), or §9 (Falsification Gates F-S8.1 through F-S8.7) is modified by this Revised entry. The principal OPEN problem F-S8.6 (H1 vs. H2 exclusion mechanism) remains the principal subject of the next revision cycle. Verification test count is unchanged at 20/20 PASS; the new p_distinct metric is reported as a parallel reading rather than a replacement criterion.